

MAE 6291

Internet of Things for Engineers

Prof. Kartik Bulusu, MAE Dept.

Week 10 [04/01/2026]

- Guest lecture: **From Silicon to Shell: The Journey of Bringing Up an IoT Platform** by **Yogesh Chobe** , Apple Inc.
- MQTT using ThingSpeak
- Using SenseHat with MQTT
- In-class Raspberry Pi Lab – ThingSpeak MQTT

git clone https://github.com/gwu-mae6291-iot/spring2025_codes.git



School of Engineering
& Applied Science

Spring 2026

THE GEORGE WASHINGTON UNIVERSITY

Photo: Kartik Bulusu

Final projects [Things to remember]



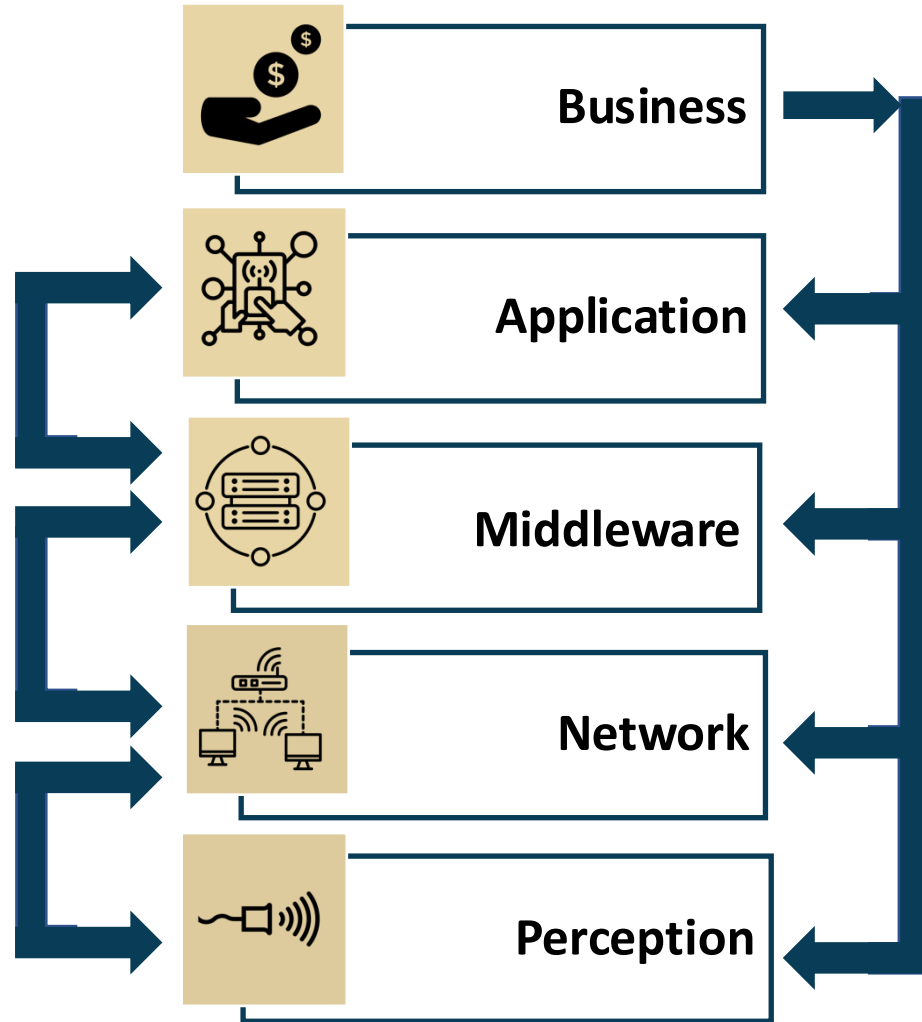


Expectations on student deliverables:

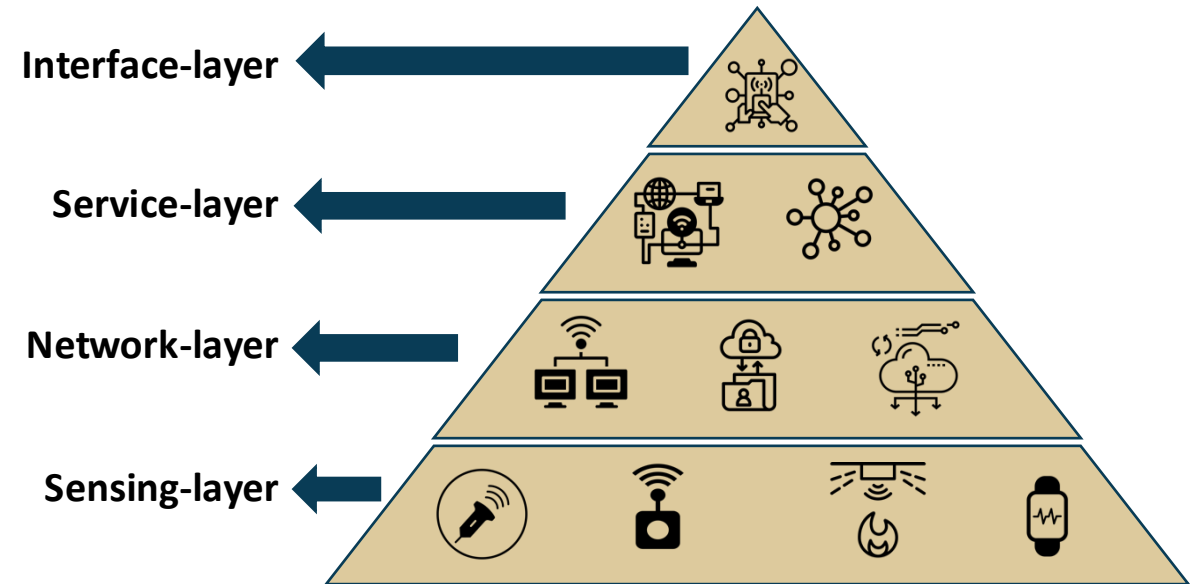
1. [Upload by 11:00 AM on April 21, 2026] Final project presentation [25 points]
2. [Upload by 11:00 AM on April 21, 2026] Final project 2-minute video [25 points]
3. [April 22, 2026] Final project demo [Tied to #1]
4. [Upload by April 21, 2026] Executive summary [10 points]
5. [Upload on April 26, 2026] Final project written report [25 points]
6. [Upload on April 29, 2026] Up to 2 Journal paper reviews by Grad students [10 points]
7. [On Google form] Share the Github repo of midterm and final projects [Tied to #1]
8. **Attendance is mandatory, no exceptions and no extensions**



The 5-Layer IoT Architecture



Service-oriented IoT Architecture



Sources:

sensor by Carolina Cani:, sensor by Pham Duy Phuong Hung, sensor by Tippawan Sookruay, sensor by Lorenzo:

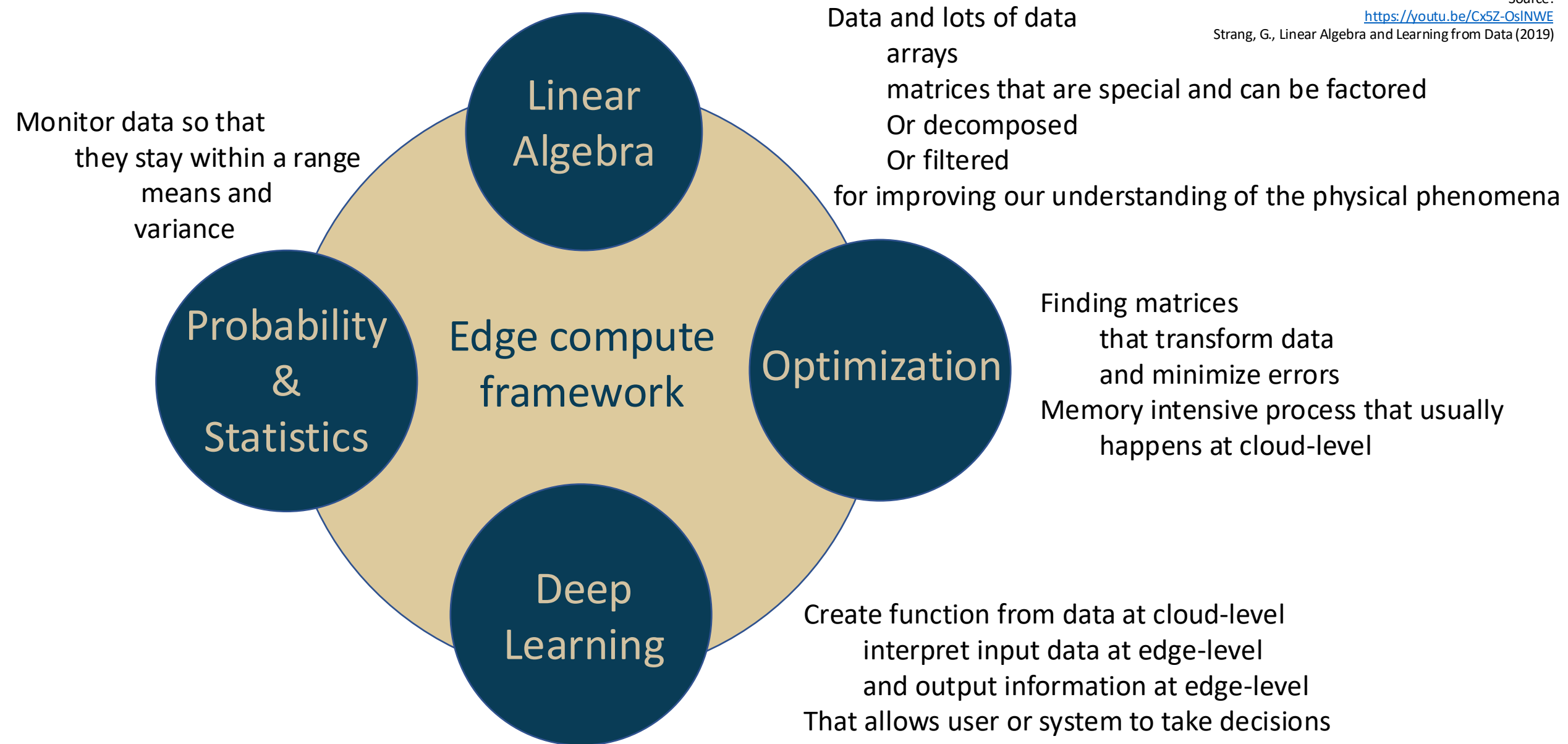
<https://thenounproject.com/browse/icons/term/sensor>

wifi network by Matthias Hartmann:: <https://thenounproject.com/browse/icons/term/wifi-network/>

application by Chaowalit Koetchuea: <https://thenounproject.com/browse/icons/term/application/>

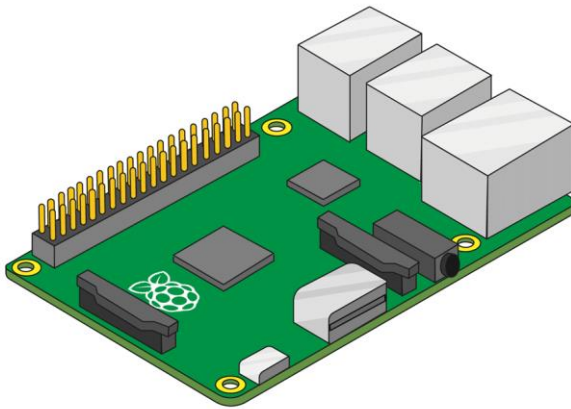
IoT Architecture layers: <https://www.startertutorials.com/blog/iot-architecture-layers.html>





Topics to be covered today

Hardware:
SenseHat



IoT Strategy #1:
Paho-MQTT with senseHat

paho-mqtt 2.0.0

```
pip install paho-mqtt
```



IoT Strategy #2:
Thingspeak with senseHat

ThingSpeak for IoT
Projects

Data collection in the cloud with advanced data
analysis using MATLAB



Explore SenseHat (The RPi companion sensor in the International Space Station)



Example of who is using the sense HAT and where
- Astro Pi



Source: https://youtu.be/kk_7KNuRLrk

What we will do today

- **Co-work**
 - Observe, ask and try in groups
- **Write small program using Python**
- **Think about**
 - Challenges, Opportunities, Gaps and Surprises

What we will learn today

- Communicate with the Sense HAT using Python
- Access the outputs of the Sense HAT
- Use the Sense HAT library to display messages and images
- Use loops to repeat certain code blocks



Step-1: “Put on the Sense HAT”

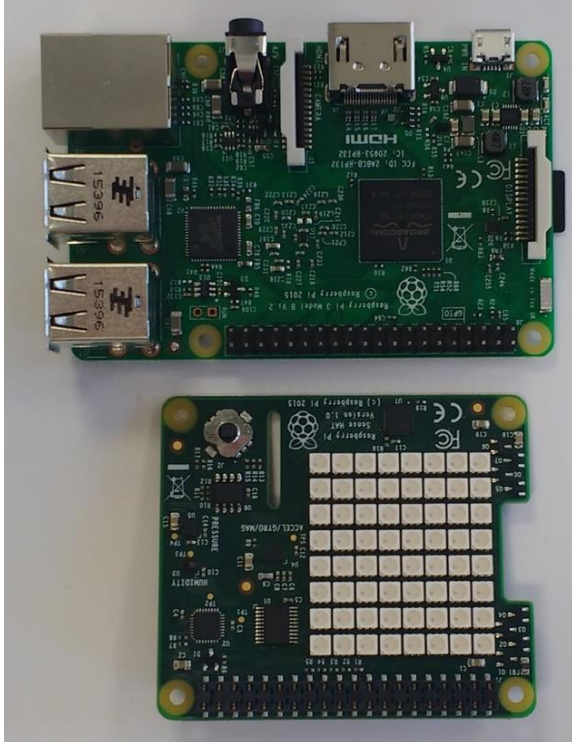


Image and animation source: <https://projects.raspberrypi.org/en/projects/getting-started-with-the-sense-hat/2>

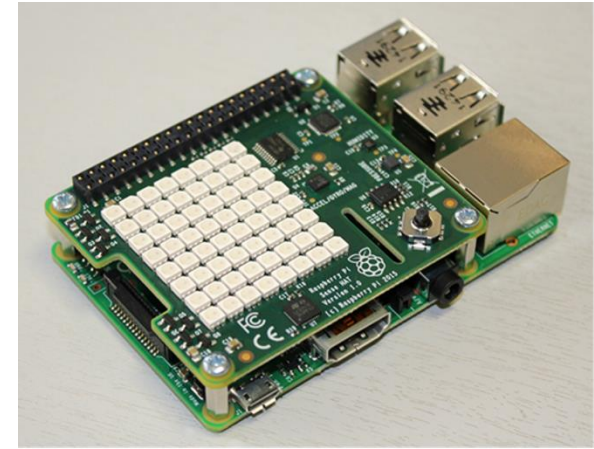
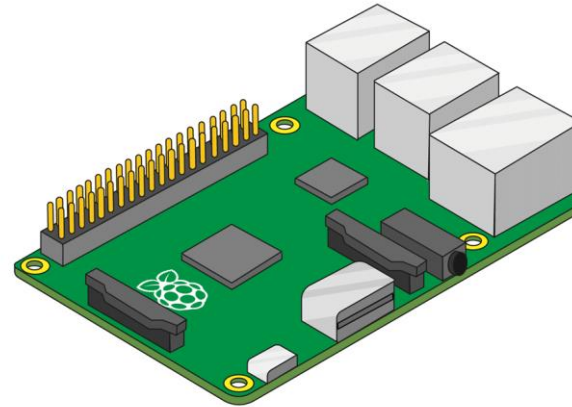


Image source: <https://reference.wolfram.com/language/ref/device/SenseHAT.html>



Source: <https://youtu.be/8NwWNOMgai4>

“and take a closer look” ...



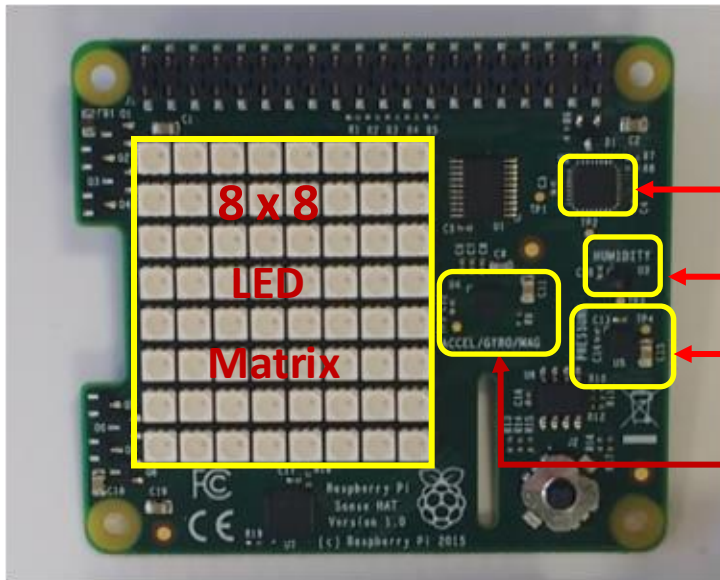
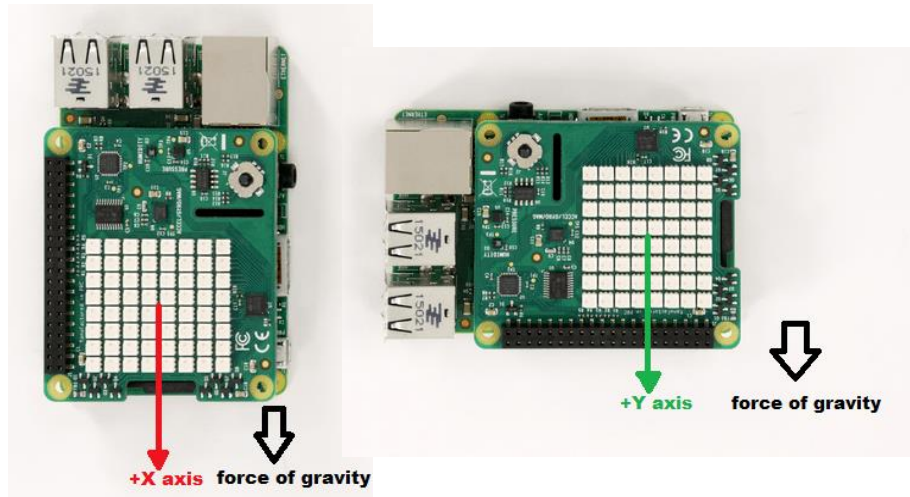


Image source: <https://projects.raspberrypi.org/en/projects/getting-started-with-the-sense-hat/2>

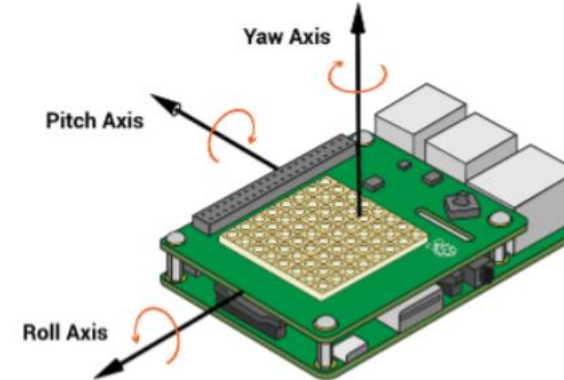


Source: <https://www.mathworks.com/help/supportpkg/raspberrypi/examples/auto-rotate-an-image-displayed-on-sense-hat-led-matrix.html>

▪ The Sense HAT has a variety of sensors that can be read from:

"Temperature"	reads temperature in degrees Celsius
"Humidity"	reads humidity in % RH
"Pressure"	reads atmospheric pressure in millibars
"Rotation"	reads gyroscopic motion in revolutions per second
"Acceleration"	reads acceleration in terms of standard accelerations due to gravity on Earth's surface
"Orientation"	reads orientation relative to magnetic north in degrees
"Magnetic Field"	reads strength and direction of a magnetic field around the sensor in microteslas

▪ The gyroscope, accelerometer, and magnetometer sensors return a list of three values that corresponds to $\{roll, pitch, yaw\}$, as oriented according to the following image:



Starting point for further exploration:

[Link for “Getting started with the Sense HAT”](#)

Source: <https://reference.wolfram.com/language/ref/device/SenseHAT.html>



Recap MQTT with senseHat Message Queuing Telemetry Transport

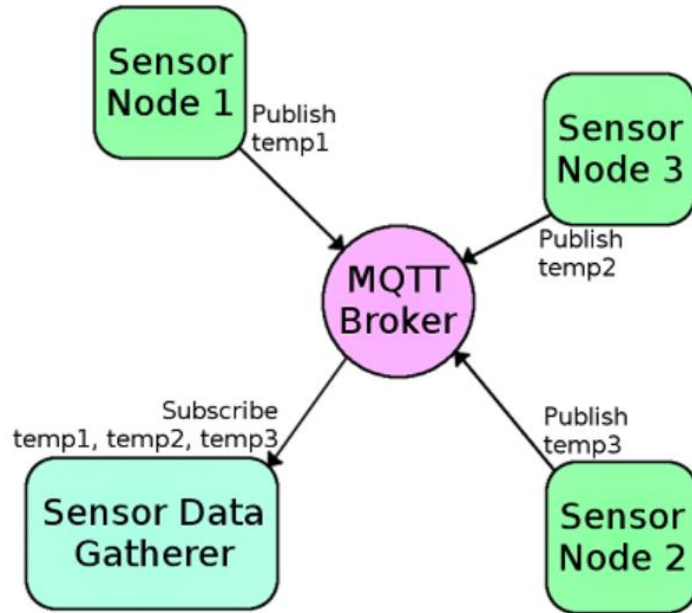
Goal: To understand how publishing and subscribing works practically



Graded in-class lab
Download codes from shared-drive and demonstrate
[10 points]



MQTT paradigm



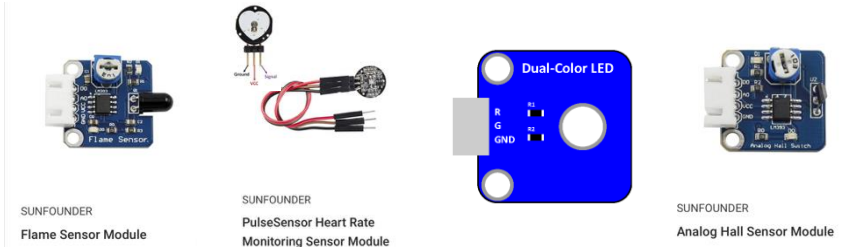
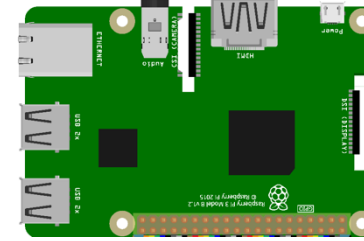
Hardware

Broker

- The broker is the server
- It distributes the information to the interested devices connected to the server.

Client

- The device that connects to broker to send or receive information.



Messaging

Topic

- The name that the message is about.
- Clients publish, subscribe, or do both to a topic.

Subscribe

- Clients tell the broker which topic(s) they're interested in.

Publish

- Clients that send information to the broker to distribute to interested clients based on the topic name.

QoS

- Quality of Service to the broker
- Integer value ranging from 0-2.



Eclipse paho - Another open source MQTT broker



paho-mqtt 1.6.1

```
pip install paho-mqtt
```



Paho is an



Eclipse-paho provides a client class which enable applications to connect to an [MQTT](#) broker to publish messages, and to subscribe to topics and receive published messages.

It also provides some helper functions to make publishing one off messages to an MQTT server very straightforward.



Step-2: Install paho-mqtt & psutil libraries

sudo apt-get update && sudo apt-get upgrade

```
pi@raspberrypi: ~  
File Edit Tabs Help  
pi@raspberrypi:~$ sudo apt-get update && sudo apt-get upgrade  
Get:1 http://raspbian.raspberrypi.org/raspbian stretch InRelease [15.0 kB]  
Hit:2 http://archive.raspberrypi.org/debian stretch InRelease  
Fetched 15.0 kB in 5s (2,647 B/s)  
Reading package lists... Done  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
Calculating upgrade... Done  
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.  
pi@raspberrypi:~$
```

sudo pip --upgrade install psutil

sudo pip install paho-mqtt

```
pi@raspberrypi: ~  
File Edit Tabs Help  
pi@raspberrypi:~$ sudo pip3 install paho-mqtt  
/usr/lib/python3/dist-packages/secretstorage/dhcrypto.py:15: CryptographyDeprecationWarning: Python 3.5 support will be dropped in the next release of cryptography. Please upgrade your Python.  
  from cryptography.utils import int_from_bytes  
DEPRECATION: Python 3.5 reached the end of its life on September 13th, 2020. Please upgrade your Python as Python 3.5 is no longer maintained. pip 21.0 will drop support for Python 3.5 in January 2021. pip 21.0 will remove support for this functionality.  
Looking in indexes: https://pypi.org/simple, https://www.piwheels.org/simple  
Requirement already satisfied: paho-mqtt in /usr/local/lib/python3.5/dist-packages (1.6.1)  
pi@raspberrypi:~$
```

sudo pip install psutil

```
pi@raspberrypi: ~  
File Edit Tabs Help  
pi@raspberrypi:~$ sudo pip3 install psutil  
/usr/lib/python3/dist-packages/secretstorage/dhcrypto.py:15: CryptographyDeprecationWarning: Python 3.5 support will be dropped in the next release of cryptography. Please upgrade your Python.  
  from cryptography.utils import int_from_bytes  
DEPRECATION: Python 3.5 reached the end of its life on September 13th, 2020. Please upgrade your Python as Python 3.5 is no longer maintained. pip 21.0 will drop support for Python 3.5 in January 2021. pip 21.0 will remove support for this functionality.  
Looking in indexes: https://pypi.org/simple, https://www.piwheels.org/simple  
Requirement already satisfied: psutil in /usr/local/lib/python3.5/dist-packages (5.9.4)  
pi@raspberrypi:~$
```

paho-mqtt 2.0.0

pip install paho-mqtt

psutil 5.9.8

pip install psutil

psutil (process and system utilities) is a cross-platform library for retrieving information on **running processes** and **system utilization** (CPU, memory, disks, network, sensors) in Python



Prof. Kartik Bulusu, MAE Dept.

Spring 2026

MAE 6291: Internet of Things for Engineers and Scientists

Explore MQTT with ThingSpeak IoT Analytics [10 points]



Thingspeak – An IoT analytics platform

ThingSpeak for IoT Projects

Data collection in the cloud with advanced data analysis using MATLAB

ThingSpeak is an IoT analytics platform service that allows you to aggregate, visualize, and analyze live data streams in the cloud.

You can send data to ThingSpeak from your devices, create instant visualization of live data, and send alerts.



Collect

Send sensor data privately to the cloud.



Analyze

Analyze and visualize your data with MATLAB.



Act

Trigger a reaction.

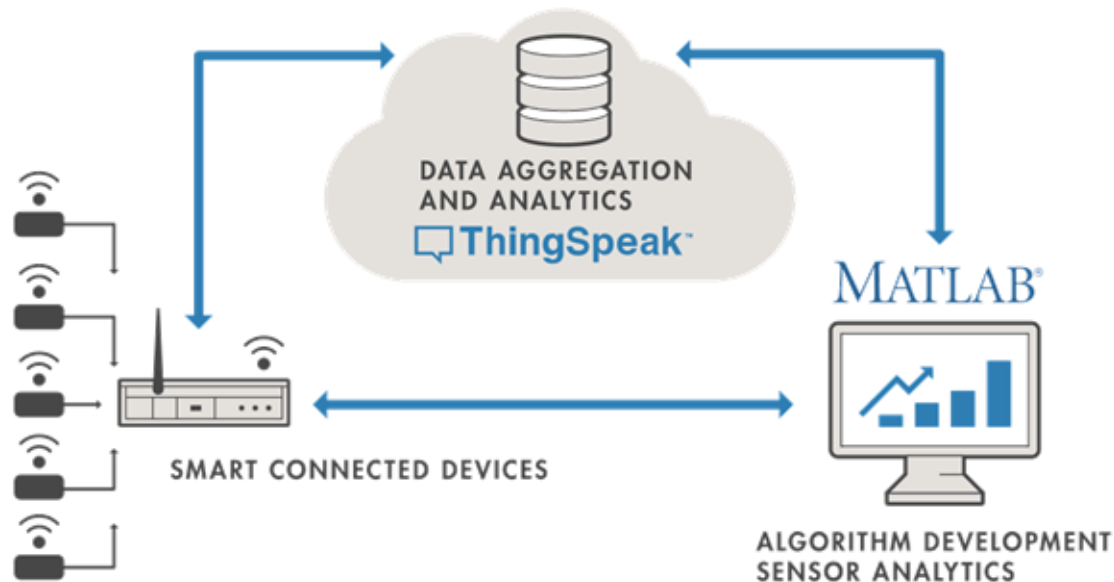


Create a Thingspeak account:

<https://thingspeak.com/login?skipSSOCheck=true>

To use ThingSpeak, you must sign in with your existing MathWorks account or create a new one.

Non-commercial users may use ThingSpeak for free.



Email



No account? [Create one!](#)

By signing in, you agree to our [privacy policy](#).

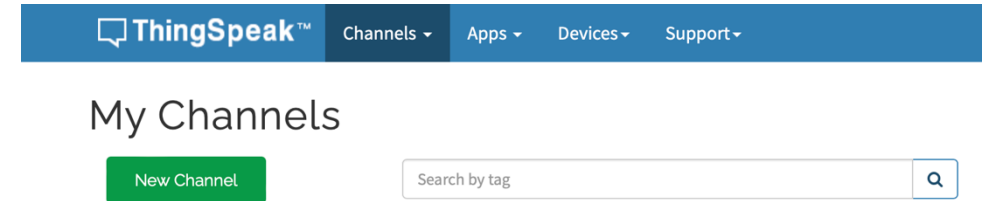
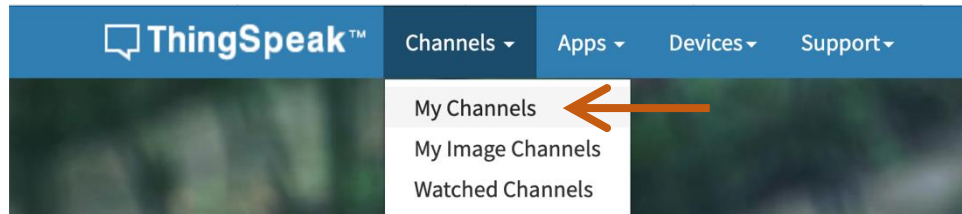
Next



Step-3: Build the ThingSpeak channels

Source:

<https://towardsdatascience.com/iot-made-easy-esp-micropython-mqtt-thingspeak-ce05eea27814>
<https://nothans.com/thingspeak-tutorials/update-a-thingspeak-channel-using-mqtt-on-a-raspberry-pi>
<https://www.mathworks.com/help/thingspeak/collect-data-in-a-new-channel.html>
<https://mapcarta.com/25030544>



Name IoT_CPU_RAM_demo

Description Demo channel for CS3907-Spring 2023

Field 1 CPU (%) ☒

Field 2 RAM (%) ☒

Field 3 ☐

Field 4 ☐

Field 5 ☐

Field 6 ☐

Field 7 ☐

Field 8 ☐

Tompkins Hall of Engineering

Type: University building
Address: 725 23rd Street Northwest, DC 20052
Categories: [building](#) and [education](#)
Location: [Washington, D.C.](#), [Mid-Atlantic](#), [United States](#), [North America](#)
[View on OpenStreetMap](#)

Latitude 38.899° or 38° 53' 56" north Longitude -77.04967° or 77° 2' 59" west

Elevation 62 feet (19 metres)

Levels 4

Operator The George Washington University

Open Location Code 87C4VXX2+H4

OpenStreetMap ID way 331159440

OpenStreetMap Feature building=university

GeoNames ID 6342433

Thanks for contributing to our open data sources. [Open data](#)



This page is based on [OpenStreetMap](#), [GeoNames](#), [Wikidata](#) and [Wikimedia Commons](#).

Tags
(Tags are comma separated)

Link to External Site

Link to GitHub

Elevation

Show Channel Location ☒

Latitude

Longitude

Show Video ☐
☒ YouTube
☐ Vimeo

Video URL

Show Status ☐

Save Channel



ThingSpeak™ Channels Apps Devices Support Commercial Use How to Buy

IoT_CPU_RAM_demo

Channel ID: 2097727 | Demo channel for CS3907-Spring 2023
 Author: kartikbulusu
 Access: Private

Private View Public View Channel Settings Sharing API Keys Data Import / Export

+ Add Visualizations + Add Widgets

Export recent data

MATLAB Analysis MATLAB Visualization

Channel 3 of 3 < >

Channel Stats

Created: about a minute ago
 Entries: 0

Field 1 Chart

IoT_CPU_RAM_demo

CPU (%)

Date

Field 2 Chart

IoT_CPU_RAM_demo

RAM (%)

Date

Channel Location

- Familiarize yourself with the panel
- Note down the Channel ID

ThingSpeak™ Channels Apps Devices Support

IoT_CPU_RAM_demo

Channel ID: 2097727 | Demo channel for CS3907-Spring 2023
 Author: kartikbulusu
 Access: Private

Private View Public View Channel Settings Sharing API Keys Data Import / Export

Channel Settings

Percentage complete 50%

Channel ID 2097727

Name IoT_CPU_RAM_demo

Description Demo channel for CS3907-Spring 2023

Field 1 CPU (%) ☒

Field 2 RAM (%) ☒



Step-4: Add MQTT device –Raspberry Pi

Source:
<https://towardsdatascience.com/iot-made-easy-esp-micropython-mqtt-thingspeak-ce05eea27814>
<https://nothans.com/thingspeak-tutorials/update-a-thingspeak-channel-using-mqtt-on-a-raspberry-pi>
<https://www.mathworks.com/help/thingspeak/collect-data-in-a-new-channel.html>
<https://mapcarta.com/25030544>

ThingSpeak™ Channels Apps Devices Support

IoT_CPU_RAM_demo

Channel ID: **2097727**
Author: **kartikbulusu**
Access: Private

Private View Public View Channel Settings Sharing API Keys Data Import / Export

MQTT

ThingSpeak™ Channels Apps Devices Support

MQTT Devices

Add a new device

Add a new device

Device Information

Name* Pi3 [Raspberry Pi 3B+] Description Pi3 to demo of ThingSpeak MQTT

Authorize channels to access

-- Select a Channel --
Your Channels & Public Channels
-- Enter a channel ID below --
Your Recent Channels
IoT_CPU_RAM_demo (2097727)
IoT1_CPU_RAM_demo (2033598)
testing_IoT2 (2097384)

Add Channel

Allow Subscribe

Cancel Add Device

New Device Added

Device Information

ThingSpeak has added a new MQTT device and authorized it to access the channels you selected.
Device Name: Pi3 [Raspberry Pi 3B+]

MQTT Credentials

Use these MQTT credentials to publish and subscribe to ThingSpeak channels. [Learn More](#)

Client ID NC0zDxwrJgEUAYxPA08Bhc
Username NC0zDxwrJgEUAYxPA08Bhc
Password

ThingSpeak does not store a copy of your device's MQTT password. Download or copy it to keep it safe.

Download Credentials
MATLAB-file (*.m)
Arduino (mqtt_secrets.h)
Plain Text (*.txt)

Done



Graded in-class lab
Download codes from shared-drive and demonstrate
[10 points]



Someone should summarize what we learned today

